

Why did direct cost per order rise to ₹115?

Eternal Ltd disclosed a **2.0% QoQ rise in Blinkit's direct cost per order to ₹115** in Q1 FY27, costing **10bps of contribution margin**. This case diagnoses which operational failure mode best explains it, using 588,832 Play Store reviews matched to the same disclosed fiscal quarters.

- Q4 FY26 → Q1 FY27
- UNSUPERVISED DISCOVERY
- TWO-PROPORTION Z-TEST
- BONFERRONI + BH
- CROSS-RUN REPLICATION
- 13-QUARTER BACK-TEST

331M	2,443	₹17,132cr	₹518
ORDERS Q1 FY27	DARK STORES +200 NET NEW	NET ORDER VALUE, Q1 FY27	AVERAGE ORDER VALUE (₹525 QOQ)

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BUILT & ANALYSED USING: [Review Diagnostic Tool](#) · [GitHub](#)

WHAT THIS CASE DIAGNOSES

₹718cr of new direct cost. Only ₹74.6cr of it is a unit-economics problem.

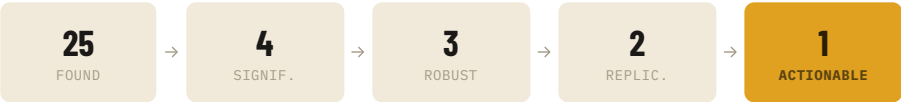
Order growth explains the rest. Direct cost per order is disclosed as a single blended figure, with nothing external splitting it into wastage, delivery or support, so the failure mode behind that ₹74.6cr has to be found rather than assumed, and then survive being tested.

VOLUME EFFECT VS RATE EFFECT



Orders grew 20.8% QoQ. The **₹74.6cr rate effect** is cost that would not have appeared had per-order efficiency simply held flat, and it is the only part worth diagnosing.

HOW THE ANSWER IS NARROWED



Categories are discovered from 588,832 reviews, then filtered by significance after multiple-comparison correction, robustness to app-version mix, and reproduction in an independent run.

WHAT WOULD COUNT AS AN ANSWER

- Moved materially in the same two quarters ✓
- Survives correcting for 25 tests at once ✓
- Not an artefact of composition or clustering ✓
- Names a mechanism touching a direct-cost line ✓

Reviews are a diagnostic signal, not a rupee measure. This case identifies and sizes a hypothesis; the causal test is handed to a controlled pilot.

DISCLOSED TRIGGER

₹115 / order
+2.0% QoQ · 10bps margin decline

SCALE OF IMPACT

₹74.6 crore
Rate effect, Q1 FY27 · derived

EVIDENCE BASE

588,832 reviews
25 complaint categories discovered

OUTCOME

1 actionable of 25
Significant, robust & replicated

Direct-cost deterioration points to cold-chain as the strongest validated operational signal worth testing.

The answer, up front. Of 25 complaint categories discovered from the corpus, one cleared every evidentiary gate and names a physical mechanism an operator can change. Most of its movement turns out to be seasonal, which lowers the headline but not the recommendation.

The conclusion

ANSWER FIRST

Cold-chain is the **strongest validated operational signal and leading root-cause hypothesis** behind the rate deterioration.

Melted and spoiled-item complaints were the only failure mode to more than double QoQ, and the only mechanism-naming signal to survive every robustness test across two independent pipeline runs.

SHARE OF NEGATIVE REVIEWS MENTIONING COLD-CHAIN ITEMS

Q4 FY26		0.90%
Q1 FY27		2.13%

RR 2.37× · P_{adj} = 2.7 × 10⁻¹⁹ · Cohen's h = 0.103
20th of 25 by prevalence · 1st by movement

THE TRIGGER IT IS BEING MATCHED AGAINST

₹115 DIRECT COST PER ORDER	+2.0% QOQ RATE RISE	₹74.6cr RATE EFFECT DERIVED
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WHERE THAT ₹74.6CR COMES FROM

₹643.8cr VOLUME EFFECT · 89.6%	₹74.6cr RATE
--	------------------------

Order growth of 20.8% explains ₹643.8cr of the ₹718cr increase. Only the ₹74.6cr rate effect is a unit-economics problem, and only that part is diagnosed here.

EVIDENCE · 01

A systematic search, not a hunch

25

→

4

→

3

→

2

→

1

25 complaint categories were discovered from 588,832 reviews without a fixed keyword list. Four moved significantly after correction, three survived reweighting for app-version mix, and two reappeared in an independent re-run.

Two of the four initial findings were eliminated by later gates. **Cold-chain is one of the two that survived**, and the only one naming a fixable mechanism.

EVIDENCE · 02

Customers describe temperature, not service

"It seems like there was no proper temperature control during delivery."

"Frozen items like icecream are not well packed."

The verbatims name packaging, temperature control and dwell time: the three things an operator can change next week. That is what separates this signal from "general dissatisfaction", which cleared the same gates but names nothing fixable.

Blinkit's 10–15 minute model has **no refrigerated transit**. Cold retention depends on dark-store storage, packaging and dwell time.

EVIDENCE · 03

Most of the movement is seasonal

≈80% SEASONAL

≈20%

A 13-quarter back-test shows the Apr–Jun peak in every year on record: 2024 at 2.23×, 2025 at 1.97×, 2026 at 2.37×. Roughly four-fifths of this quarter's rise follows the recurring summer pattern.

The **residual above historical seasonality** is the portion worth investigating operationally, and it sits on top of a penalty that repeats every April.

The recommendation

LOWEST-REGRET FIRST

1 Insulated liners

₹3–9 gel-pack insert at pack-out. Reversible, testable in one season.

2 Freezer audit

Diagnostic. Tests whether the failure sits upstream of packaging.

3 Dwell-time SLA

Systemic. Higher engineering effort, guarded by on-time delivery.

Test the lowest-regret intervention with a controlled pilot: matched treated and control dark stores, weekly complaint-share reads, stop rules declared in advance, rather than committing to a rollout on observational evidence.

Confidence boundary

READ THIS WITH THE ABOVE

Observed review data identifies the signal. Internal cost data and the pilot are required to establish financial causality.

This case does **not** claim cold-chain caused the ₹74.6cr rate effect. It claims cold-chain is the strongest signal that survived testing, sizes it under a stated assumption, and specifies the experiment that would settle it.

Evidence tiers used throughout this deck. Disclosed: ₹115 direct cost/order, +2.0% QoQ, 10bps contribution-margin decline (Eternal Ltd, Q1 FY27). Derived: ₹74.6cr rate effect and the volume/rate decomposition. Measured: complaint-share movement from own pipeline. Tested: significance, version-mix robustness, independent replication, 13-quarter seasonality. Assumed: 25% cold-chain attribution and 10% frozen-order share, both registered in section 10. Hypothesized: the thermal / packaging / dwell-time mechanism in section 09.

Direct cost rose ₹718cr in one quarter. Only ₹74.6cr of it is an efficiency loss. That is the part worth diagnosing.

Blinkit grew orders 20.8% QoQ, so most of the cost increase is simply volume. Stripping that out isolates the **rate effect**: the ₹2.25 per order that Eternal disclosed as a 2.0% QoQ rise, and the 10bps of contribution margin it consumed.

Decomposing the ₹718cr increase

VOLUME VS RATE EFFECT

TOTAL DIRECT-COST INCREASE, Q4 FY26 → Q1 FY27

₹718.4cr

₹643.8cr

89.6% • VOLUME EFFECT

₹74.6cr

10.4% • RATE

57.1M more orders × ₹112.75 old rate

331M orders × ₹2.25 rate rise

Growth explains nearly nine-tenths of the increase and is a deliberate strategic cost. The remaining **₹74.6cr is unit-economics deterioration**: cost that would not have appeared had per-order efficiency simply held flat.

THE ARITHMETIC BEHIND THE RATE EFFECT

₹112.75

Q4 FY26
IMPLIED

+

₹2.25

RATE RISE
+2.0% QOQ

=

₹115.00

Q1 FY27
DISCLOSED

×

331M

ORDERS
Q1 FY27

THE DIAGNOSTIC QUESTION

Which operational failure mode best explains the ₹2.25 per-order rate deterioration, and is it large enough to justify action?

What Eternal disclosed

BLINKIT SEGMENT

METRIC	Q4 FY26	Q1 FY27	Δ QOQ
Direct cost / order the disclosed trigger	₹112.75 implied	₹115.00	+2.0%
Contribution margin level not consistently reported	n/a	n/a	-10bps
Net order value	₹14,386cr	₹17,132cr	+19.1%
Orders	273.9M	331.0M	+20.8%
Average order value	₹525	₹518	-1.3%
Take rate	26.9%	27.5%	+60bps
Adjusted EBITDA	₹37cr	₹102cr	+176%
Total direct cost orders × cost/order	₹3,088cr	₹3,806cr	+23.3%

Q4 FY26 cost per order is not separately disclosed, so ₹112.75 is implied by reversing the reported +2.0% QoQ. Net order value cross-checks against orders × AOV to within 0.1% in both quarters. Contribution-margin **level** is reported inconsistently, so only the 10bps delta is used.

WHY 10BPS MATTERS

Take rate **improved 60bps** while direct cost per order **worsened 2.0%**. Blinkit monetised better and executed worse in the same quarter. With the segment only just adjusted-EBITDA positive, the slip is an operations problem, not a pricing one.

FOR SCALE

₹74.6cr is **2.0%** of Q1 FY27 total direct cost and **73%** of the segment's ₹102cr adjusted EBITDA. Held at this run-rate it is ₹298cr a year.

WHAT "DIRECT COST PER ORDER" CAN CONTAIN

Cost lines a complaint category could plausibly map to:

Last-mile delivery

Dark-store operations

Inventory wastage

Packaging

Replacements & refunds

Customer support

WHY THIS IS HARD TO DIAGNOSE EXTERNALLY

Eternal discloses direct cost per order as a **single blended number**, with no wastage, delivery or support split. An independent signal is needed to attribute the movement to any one operational cause.

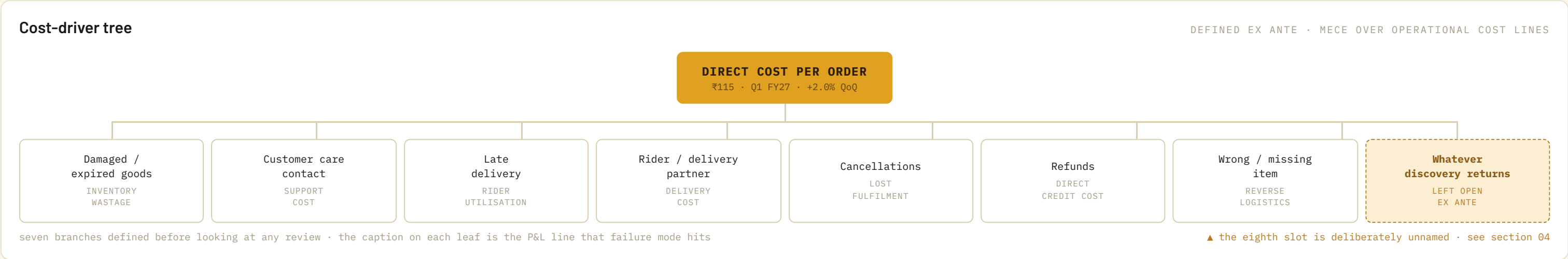
WHAT WOULD COUNT AS AN ANSWER

A failure mode that **moved materially** in the same quarter, survives correction for multiple testing, is not explained by composition, and **names a mechanism** that plausibly touches a direct-cost line.

Sources: Eternal Ltd Q1 FY27 results (direct cost/order ₹115, +2.0% QoQ, 10bps contribution-margin decline). Order volume, AOV and dark-store counts from Q1 FY27 investor-deck coverage (Business Standard, Lapaas Voice, INDmoney), Jul-Aug 2026. **Derived by author:** implied Q4 FY26 cost/order, implied GOV, total direct cost, and the volume/rate decomposition.

Seven operational failure modes were written down before any review was read. The eighth had to be discovered.

The branches below cover every cost line an operational failure can hit, without overlapping each other, and all seven were defined **ex ante**. Complaint categories were then discovered from the corpus rather than matched to this list, so the search could return a failure mode nobody had thought to look for.



25 failure modes emerged from the corpus. The biggest complaint category was not the one that moved.

Named by the model from its own top terms, then labelled in plain English. **Prevalence and change are different questions**: the largest category barely moved, and the one that doubled sits 20th by size. Showing all 25 is what demonstrates the finding was not cherry-picked.

All 25 discovered categories			NEGATIVE BAND · RANKED BY Q1 FY27 SHARE		
#	CATEGORY	MODEL-GENERATED TOP TERMS	Q4 FY26	Q1 FY27 ← DEC · QOQ · INC →	QOQ
01	Delivery time & location	delivery, time, late, location	15.05%	14.75%	0.98×
02	General dissatisfaction ●	blinkit, experience blinkit, order, money	9.69%	11.17%	1.15×
03	App usability	app, worst app, worst, dont	10.03%	10.67%	1.06×
04	Hindi / Hinglish cluster	hai, bhi, nhi, se	9.73%	10.33%	1.06×
05	Customer support access	customer, support, customer support, care	7.00%	7.62%	1.09×
06	Missing / wrong items	missing, order, items, item	7.26%	7.25%	1.00×
07	Refunds & returns	refund, return, product, damaged	7.01%	7.20%	1.03×
08	Delivery & handling charges ●	charges, charge, delivery charges, handling	8.01%	6.72%	0.84×
09	Service quality (generic)	service, worst service, bad service	5.80%	5.90%	1.02×
10	Product quality (generic)	quality, product, products, bad	3.97%	4.04%	1.02×
11	Pricing / too expensive	price, high, price high, prices	3.76%	3.63%	0.96×
12	Produce freshness (rotten) ●	vegetables, rotten, fruits, fresh	3.03%	3.62%	1.20×
13	Frustration language	frustrating, disappointed, unacceptable	3.44%	3.48%	1.01×
14	Packaging & bags	bag, packaging, packing, dirty	3.02%	3.48%	1.15×
15	Payment & UPI failures	payment, upi, deducted, order	3.59%	3.42%	0.95×
16	Mixed good / bad	good, good good, bad, bad bad	2.95%	2.89%	0.98×
17	Order cancellation	cancel, cancel order, order, cancellation	2.96%	2.88%	0.97×
18	Bad experience (generic)	experience, experience bad, bad experience	2.37%	2.69%	1.13×
19	Expired products	expired, expiry, date, expired products	2.11%	2.23%	1.06×
20	Cold-chain / melted items ●	ice, ice cream, cream, melted	0.90%	2.13%	2.37×
21	Coupons & offers	coupon, offers, discount, offer	2.12%	2.04%	0.96×
22	Cash on delivery	cash, cash delivery, delivery, delivery option	2.01%	1.95%	0.97×
23	Chatbot / AI support	chat, support, bot, ai	1.95%	1.94%	1.00×
24	Competitor comparison	zepto, better, instamart, flipkart	1.88%	1.62%	0.86×
25	Surge charges	surge, surge charge, charge, surge charges	1.48%	1.35%	0.91×

The eighth branch, filled in		EX ANTE TREE VS DISCOVERY	
BRANCH DEFINED EX ANTE	CATEGORY DISCOVERY RETURNED	Q1 SHARE	QOQ
Damaged / expired	expired, expiry, date	2.23%	1.06×
	vegetables, rotten, fruits	3.62%	1.20×
Customer care contact	customer, support, care	7.62%	1.09×
Late delivery + rider	delivery, time, late, partner	14.75%	0.98×
Cancellations	cancel, cancel order	2.88%	0.97×
Refunds	refund, return, policy	7.20%	1.03×
Wrong / missing item	missing, items, delivered	7.25%	1.00×
Unnamed eighth slot	ice, ice cream, melted, chilled	2.13%	2.37×

WHERE A FIXED LIST WOULD HAVE FAILED

Lumping melted ice cream into **damaged / expired** shows a 1.06× move. Split out on its own it is 2.37×. The signal only appears at the right level of detail.

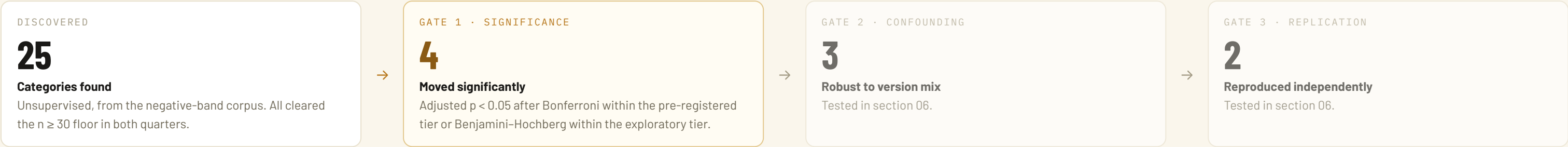
What the output tells us		READING THE TABLE
"Hindi / Hinglish", "App usability" and "Frustration language" are linguistic groupings, not failure modes. No cost claim rests on them.		
01	Size and movement are unrelated	Delivery time is largest at 14.75% and moved 0.98×. Cold-chain is 20th at 2.13% and moved 2.37×. Ranking by volume finds nothing.
02	Discovery split what a keyword list would merge	Produce freshness, expired products, packaging and cold-chain separated into four distinct failure modes that one "damaged goods" tag would have averaged away.
03	Speed is a strength, not the problem	The same pipeline on 4–5★ reviews returns one dominant praise theme: <i>fast delivery, quick, on time</i> (42,960 snippets).

Almost every change bar sits on the centre line. **Only one category moved far enough to be worth testing.** Whether it survives that testing is the next section.

● = significant after multiple-comparison correction (4 of 25). Change bar scaled to |QoQ - 1| with a maximum of 1.40. A review may match more than one category, so shares do not sum to 100%. **Source:** own BERTopic discovery over the Q4 FY26 + Q1 FY27 negative-review window (52,664 clause snippets from 35,325 reviews). Shares are the percentage of each quarter's negative reviews matched to a category by cosine similarity against category centroids, using per-category calibrated thresholds. Q4 FY26 n = 15,885. Q1 FY27 n = 19,440.

Of 25 categories, four moved significantly. The other 21 are genuine nulls, not underpowered tests.

Testing 25 categories at once inflates the chance of a false positive, so every p-value is corrected before anything is carried forward. The more important question is the reverse one: whether the study could have *seen* a real change in the categories that stayed flat. It could.



The four categories that cleared Gate 1

Q4 FY26 → Q1 FY27 · NEGATIVE BAND

CATEGORY	Q4 FY26	Q1 FY27	QOQ	COHEN'S H	P (RAW)	P (ADJUSTED)	TIER	GATE 1
Cold-chain / melted items ice, ice cream, cream, melted, icecream	0.90%	2.13%	2.37× ▲	0.103	2.1×10^{-20}	2.7×10^{-19}	Exploratory	PASSES
General dissatisfaction blinkit, experience blinkit, order, money	9.69%	11.17%	1.15× ▲	0.048	7.1×10^{-6}	8.5×10^{-5}	Pre-registered	PASSES
Produce freshness (rotten) vegetables, rotten, fruits, fresh, quality	3.03%	3.62%	1.20× ▲	0.033	2.1×10^{-3}	2.5×10^{-2}	Pre-registered	PASSES
Delivery & handling charges charges, charge, delivery charges, handling	8.01%	6.72%	0.84× ▼	0.049	3.5×10^{-6}	2.3×10^{-5}	Exploratory	PASSES

Cohen's h is reported as absolute magnitude with direction shown by the arrow. Cold-chain's adjusted p-value is fourteen orders of magnitude below the next-nearest candidate, and it is the only one of the four whose movement exceeds a doubling. The remaining 21 categories are listed in full in section 04; none reached the corrected threshold.

WHY THE 21 FLAT CATEGORIES ARE INFORMATIVE

Delivery, support, refunds, cancellations

The four operational branches most managers would reach for first. None showed a statistically significant increase, and at their prevalence, a 10–15% move would have been detected.

A null is not a "flat" claim

The correct statement is **no statistically significant increase was detected**, at a detection floor documented opposite, not that these categories are unchanged in the business.

This is what eliminates alternatives

Every one of these is a competing explanation for the rate rise. Their non-detection is what narrows the field, and it is why the recommendation in section 11 excludes rider-speed and support programmes.

Power • could we have seen it?

MDE

Minimum detectable Cohen's h = **0.040** at 80% power, $\alpha = 0.05$, Bonferroni-corrected, 12 branches.

BASELINE SHARE	DETECTABLE FROM	REL .
2% e.g. cold-chain	2.59%	+30%
7% e.g. refunds	8.05%	+15%
15% e.g. delivery	16.44%	+10%

At a 2% baseline the study could detect roughly a +30% relative movement at 80% power. Cold-chain moved +137%. This was never a close call, and the 21 categories that did not move are genuine nulls, not underpowered tests.

WHAT EACH CHECK ASKS

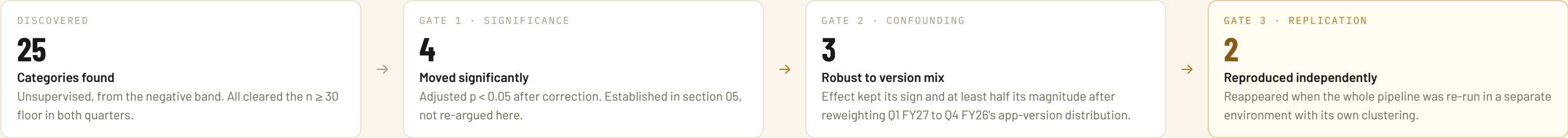
Discover BERTopic	What failure modes exist at all?
Test two-proportion z	Which categories' share actually changed?
Correct Bonferroni / BH	Which survive testing 25 things at once?
Power MDE, Cohen's h	Could the study even see a real change?

Significance is the first of three gates, and the one that eliminated the fewest candidates.

Statistics: two-proportion z-test on category share of negative reviews, Q4 FY26 (n = 15,885) vs Q1 FY27 (n = 19,440). Adjusted p-values use Bonferroni within the pre-registered tier and Benjamini–Hochberg within the exploratory tier; tier assignment was fixed before testing. **Power:** minimum detectable effect computed at 80% power, two-sided $\alpha = 0.05$, Bonferroni-corrected across the 12 pre-registered branches. Full per-category statistics are retained in the supporting material.

Two of the four significant findings did not survive. Composition removed one; independent replication removed another.

A significant p-value is not enough on its own. Two further gates ask whether the movement is an artefact of **who was using which app version**, and whether it reappears when the entire pipeline is re-executed in a separate environment with its own clustering. Both were applied before any candidate was carried into the business case.



Every candidate against every gate

CATEGORY

QOQ SIGNIFICANCE

VERSION MIX

REPLICATION

VERDICT

Cold-chain / melted items ice, ice cream, cream, melted	2.37× ▲	✓	✓	✓	PRIORITY
General dissatisfaction blinkit, experience blinkit, order, money	1.15× ▲	✓	✓	✓	CORROBORATING
Produce freshness (rotten) vegetables, rotten, fruits, fresh	1.20× ▲	✓	✓	×	NOT CARRIED
Delivery & handling charges charges, charge, delivery charges	0.84× ▼	✓	×	×	NOT CARRIED

GATE 2 DETAIL • effect size in percentage points, before and after reweighting Q1 FY27 to Q4 FY26's app-version mix

Cold-chain

raw +1.23pp

rewtd +1.39pp

ROBUST

Sign held and magnitude grew. The rise is not an artefact of which app version users were on.

→ Carried to Gate 3.

General dissatisfaction

raw +1.47pp

rewtd +1.96pp

ROBUST

Sign held and magnitude grew. Survives, but names no single fixable mechanism.

→ Carried to Gate 3.

Produce freshness

raw +0.59pp

rewtd +0.87pp

ROBUST

Passed this gate, then failed replication in the independent re-run (adjusted $p = 0.065$).

→ Dropped at Gate 3.

Delivery & handling

raw -1.29pp

rewtd -0.32pp

NOT ROBUST

Three-quarters of the decline disappeared on reweighting. Mostly app-version mix, not sentiment.

→ Dropped at Gate 2.

Independent replication

CATEGORY

RUN A

RUN B

RESULT

Cold-chain / melted items	2.68× sig · robust	2.37× sig · robust	REPLICATES
General dissatisfaction	1.14× sig · robust	1.15× sig · robust	REPLICATES
Produce freshness (rotten)	1.18× not significant	1.20× sig · robust	CONTRADICTS
Delivery & handling charges	0.86× sig · not robust	0.84× sig · not robust	WEAK IN BOTH

REVIEW COUNTS

Identical

15,885 / 19,440

CATEGORIES FOUND

22 vs 25

by environment

SURVIVED BOTH

2 of 4

1st in both runs

WHY REPLICATION IS TREATED AS A GATE

Clustering varied across environments, returning 22 categories in one run and 25 in the other, while **the data layer remained identical**: byte-identical cohorts from the same scrape. Every discrepancy originates in clustering, nowhere else, so a finding that depends on one clustering is treated as unproven. The resolution itself was selected against **pre-defined separation criteria**, not tuned to the answer. Full discrepancy analysis in the supporting material.

Findings that appeared in only **one** of the two runs, packaging & milk damage and app usability, were **discarded rather than reported**. That discipline is why only two of four candidates carried forward.

Gate 2: direct-standardisation reweighting on app-version groups, requiring $\geq 50\%$ version-mix coverage; a finding is robust if it keeps its sign and at least half its magnitude. **Gate 3:** full pipeline re-executed independently in a separate environment. **Run A** = local execution, 22 negative categories found. **Run B** = independent cloud execution, 25 categories found. Both runs applied to identical review counts drawn from the same scrape; significance and robustness assessed independently within each run, using that run's own category family for the multiple-comparison correction.

Cold-chain is the strongest validated operational signal. Customers describe a temperature problem, not a service problem.

One category emerged from the funnel with a mechanism attached. This section is about what that signal actually says and what can be done with it. The evidence that it survived every gate is already established in sections 05 and 06 and is not re-argued here.

Movement

Q Q Q

2.37×

RISE IN SHARE OF NEGATIVE REVIEWS

Q4 FY26

0.90%

Q1 FY27

2.13%

Complaint mentions, Q4 FY26

143

Complaint mentions, Q1 FY27

415

Rank by prevalence (of 25)

20th

Rank by movement

1st

TRIANGULATION ACROSS RATING BANDS

Neg 1–2★

2.13%

Neutral 3★

1.09%

Pos 4–5★

0.26%

The theme concentrates sharply in the negative band, consistent with a genuine failure rather than a topic customers simply mention more often.

! THE CLUSTER IS NOT PERFECTLY CLEAN

Random sampling found unrelated complaints inside it. The signal is real, but **2.13% is an upper bound** on pure cold-chain mentions.

Customer voice

VERBATIM • Q1 FY27

"It seems like there was no proper temperature control during delivery."

"The upper cover of the ice cream was also missing, and it was in a torn and melted condition."

"Frozen items like icecream are not well packed."

"at last i received only the ice cream juice"

"After sometime we found the the tub was broken from bottom."

"one packet of milk was damaged, I returned and he promised to bring new one."

"Fast delivery means nothing if basic quality standards are ignored."

"received 3 ice-cream instead of 4."

⚡ audited: missing item, not a cold-chain failure

"And the customer care is just a joke."

⚡ audited: support complaint, not a cold-chain failure

All nine drawn at random from the 601 Q1 FY27 reviews in this category, **not hand-picked**. They are representative, not selected. Representative-document sampling was avoided because it favours centroid-proximal text and hides minority sub-themes. The last two are shown deliberately: **they are the contamination**, quantified rather than quietly removed.

Actionability

VS THE OTHER SURVIVOR

General dissatisfaction cleared the same three gates. It is retained as corroboration, but it names nothing an operator can change. Cold-chain names three things:

LEVER 01

Packaging

Insulation at pack-out

LEVER 02

Temperature

Dark-store cold storage

LEVER 03

Dwell time

Exposure before dispatch

01 It maps to a specific cost line

A melted unit cannot be resold. That points at inventory wastage, a replacement trip, or a refund, all inside direct cost per order.

02 It has a physical explanation

A 10–15 minute model has no refrigerated transit. Cold retention depends entirely on dark-store storage, packaging and dwell time before pickup.

03 It behaves like a real-world effect

The signal follows the weather across three years and concentrates in the negative band: patterns you would expect from a thermal failure, not from a shift in how people write reviews.

Evidence boundary

WHAT THIS SLIDE CLAIMS

OBSERVED

Complaint mentions rose 2.37×, replicated.

HYPOTHESIZED

The thermal / packaging / dwell-time failure mechanism.

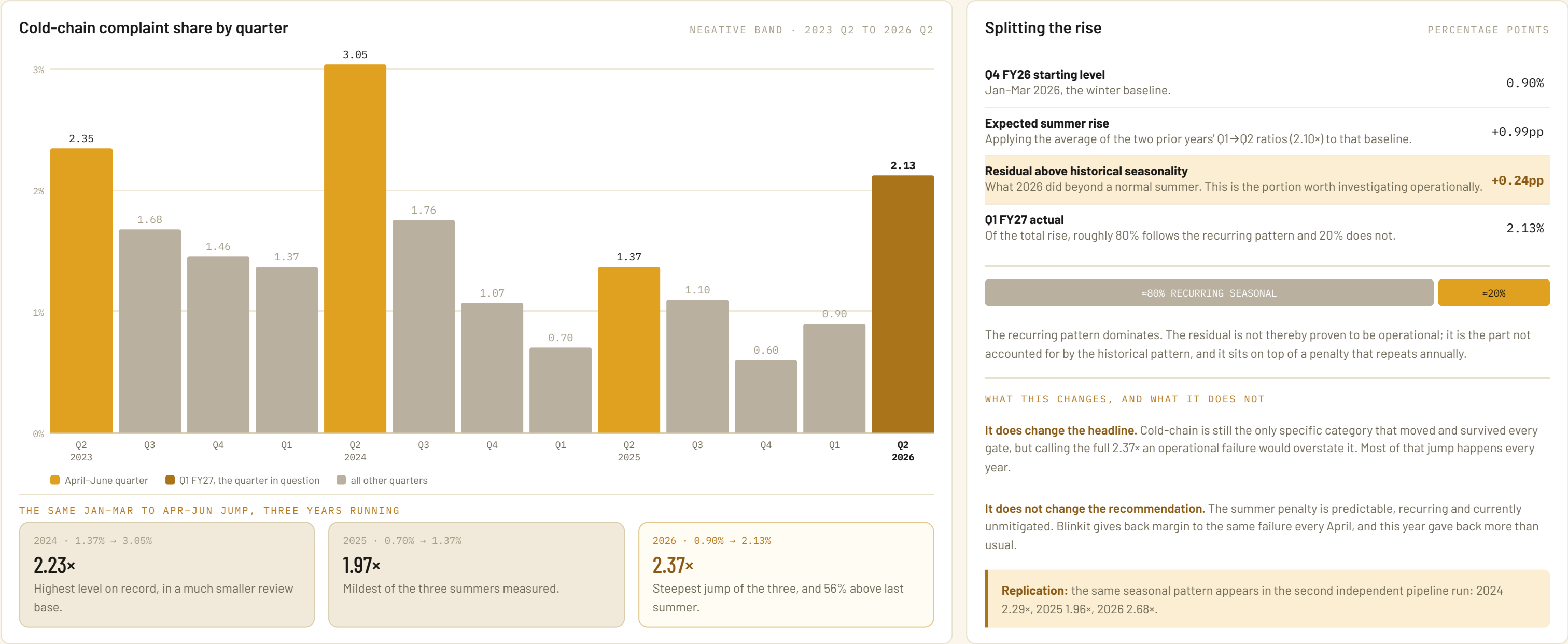
NOT MEASURED

Direct financial attribution to the rate effect.

Category: discovered cluster with model-generated terms *ice*, *ice cream*, *cream*, *melted*, *icecream*, *cold*, *chilled*. 143 of 15,885 Q4 FY26 negative reviews and 415 of 19,440 Q1 FY27 negative reviews matched. **Quotes:** randomly sampled from the 601 Q1 FY27 reviews assigned to the category via cosine similarity, filtered only for length. Band shares from the same category centroid applied to the Neutral (3★) and Positive (4–5★) discovery runs. Statistical evidence for this category is set out in sections 05 and 06 and is not restated here.

Cold-chain complaints spike every April. Most of the rise is a recurring seasonal pattern, not new deterioration.

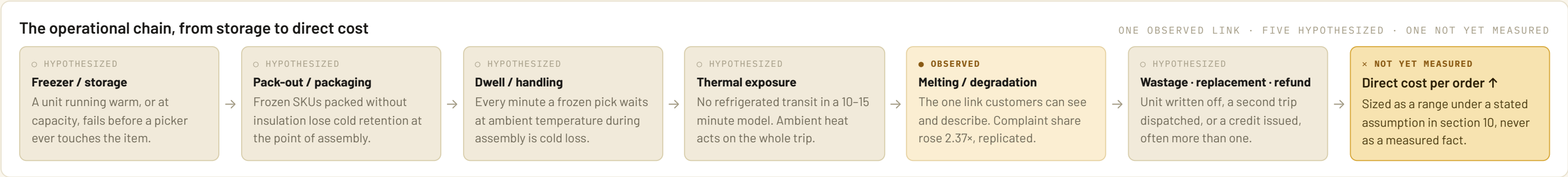
Extending the same measurement back three years challenges the headline. The Apr–Jun peak has happened in every year on record, so roughly **four-fifths of this quarter's rise is the normal summer pattern**. Only the residual above that pattern is the portion worth investigating operationally.



Method: the Q1 FY27 category definition was applied unchanged to every prior quarter in the corpus, so the series measures one consistent definition over time rather than re-clustering each period. Quarters with fewer than 3,000 negative reviews are excluded, as is the Jul 2026 partial quarter. **Note:** review volume roughly doubled across the series, which is why shares rather than counts are compared. The 2.10× expected ratio is the mean of the 2024 and 2025 Q1–Q2 ratios only; with three observations it is an estimate, not an established constant.

Root-cause analysis: where can cold-chain failure enter the operation, and how would it reach a cost line?

This is where the case moves from review analytics to operations diagnosis. The chain below is the plausible operational path from a customer's complaint to a line on the P&L. It is **a hypothesis with one observed link**, not a confirmed accounting of it. Each stage names the internal data that would confirm it.



₹74.6cr is a derived number. Everything past it is a sensitivity range on a stated assumption.

The rate effect is arithmetic, not estimation. What cold-chain is *worth* depends entirely on how much of that ₹74.6cr this one mechanism explains, which review data alone cannot tell us. The range below is deliberately wide, and **25% is a working assumption, not a measurement**. The question this slide answers is only whether the remaining problem is large enough to be worth testing.

The defensible base

DERIVED

₹74.6cr

331M orders × ₹2.25 rate increase.
Deterministic arithmetic from disclosed figures, established in section 02.

Of Q1 FY27 total direct cost	2.0%
Of Q1 FY27 adjusted EBITDA	73%
Annualised at this run-rate	₹298cr

Carries no attribution. This is the size of the whole rate problem, before asking which failure mode caused any of it.

Attribution sensitivity

ASSUMED RANGE, NOT MEASURED

If cold-chain explains X% of the ₹74.6cr rate effect, the quarterly and annualised opportunity is:

10%

₹7.5cr · ₹30cr p.a.

25%

₹18.7cr · ₹75cr p.a.

50%

₹37.3cr · ₹149cr p.a.

25% is a working assumption, not a measured fact:

a mid-point judgement bounded roughly ±2× either way, and the single largest source of uncertainty in this case. Cold-chain is not claimed to be the *only* contributor to the rate rise, only the one this analysis could validate.

Illustrative opportunity

25% ANCHOR

₹18.7cr

Illustrative only. Moves with the attribution assumption; not a realized or recoverable figure.

Of adj. EBITDA	18%
Annualised	₹75cr

FOR SCALE

Blinkit's entire Q1 FY27 adjusted EBITDA was ₹102cr. Even the conservative end of this range is material against that base, which is why it warrants a controlled test.

Assumption register

EVERY INPUT, CLASSIFIED

ITEM	STATUS
₹115 direct cost / order Eternal Q1 FY27 disclosure	DISCLOSED
+2.0% QoQ rate rise Eternal Q1 FY27 disclosure	DISCLOSED
331M orders, ₹518 AOV investor-deck coverage, cross-checked	REPORTED
₹74.6cr rate effect deterministic arithmetic	DERIVED
Cold-chain complaint rise 2.37× own pipeline, replicated	MEASURED
Seasonality ≈80% of the rise 13-quarter back-test, section 08	TESTED
Thermal / packaging / dwell mechanism section 09, not directly observed	HYPOTHEZIZED

25% cold-chain attribution
judgement, bounded 10–50%

ASSUMED

10% of orders carry frozen items
illustrative only, not sourced

ASSUMED

The ₹5.65 conditional spend threshold, and how it is built

BREAK-EVEN PER FROZEN / CHILLED ORDER

CONDITIONAL THRESHOLD =

Attributable opportunity ₹18.7cr

331M orders × assumed frozen share 10%

=

₹5.65

Both inputs are assumptions: the 25% anchor sets the numerator, the assumed 10% frozen/chilled share the denominator. Change either and the threshold moves. It is a conditional budget, not an observed cost figure.

IF FROZEN / CHILLED ORDERS ARE...	5%	10% (ASSUMED)	15%	20%	ALL 331M ORDERS
Conditional budget per such order	₹11.30	₹5.65	₹3.77	₹2.82	₹0.56

WHAT AN INTERVENTION ACTUALLY COSTS

Gel-pack bulk pricing (IndiaMART, Aug 2026) at 1,000–5,000 piece MOQs:

Blucia Overseas		₹3
Shakya World Trade		₹6
Valor Industries		₹9

The lower half of the ₹3–9 range clears the ₹5.65 threshold; the upper half does not. That is precisely why the pilot's first gate is a procurement quote, before any rollout spend is committed.

READ THIS TABLE BEFORE THE NUMBERS

The two red rows are the only places where judgement enters the sizing. Everything above them is disclosed, derived, measured or tested. ₹18.7cr and ₹5.65 both sit downstream of those two rows and should never be quoted without them.

WHAT WOULD REPLACE THE ASSUMPTIONS

Internal wastage, replacement and refund cost attributable to frozen and chilled SKUs, plus the actual frozen share of the basket mix, both available inside Blinkit, neither available externally. The pilot in section 12 is designed to surface exactly these two figures.

Break-even table assumes the 25% attribution anchor throughout; each column divides the resulting attributable opportunity by 331M orders × the stated frozen share. Gel-pack prices from IndiaMART bulk listings (Aug 2026): Blucia Overseas ₹3/piece, Shakya World Trade ₹6/piece, Valor Industries ₹9/piece, all at 1,000–5,000 piece minimum order quantities. Reusable insulated bags (~₹944/unit, TradeIndia) are a separate amortised capex item and are not included in the per-order comparison.

Test the cold-chain hypothesis with the lowest-regret intervention.

Three interventions, each aimed at a specific link in the hypothesized mechanism, ordered by cost-to-test and directness rather than by assumed impact. None requires internal data to begin, and none is recommended for rollout ahead of the controlled test in section 12.

1 · Insulated liners & gel packs

DO FIRST

Mandated at dark-store pack-out for all frozen and dairy SKUs. Interrupts the mechanism at its **earliest and cheapest** point, before the item ever leaves the store.

WHICH LINK IT INTERRUPTS

Pack-out and thermal exposure. If cold retention holds, none of the downstream write-off, replacement or refund costs are triggered.

Conditional threshold illustrative	₹5.65
Bulk gel-pack price	₹3 – ₹9
Time to first read	6 weeks

PRIMARY METRIC

Weekly cold-chain complaint share, target back **under 1.2%** within one quarter (from 2.13%).

WHY THIS ONE FIRST

Lowest effort, most direct, and fully reversible. It is also the only one of the three that can be tested end-to-end inside a single season, and the packaging link is the one customers name explicitly.

Timing matters. The spike is seasonal and arrives every April. Liners need to be in stores **before** the season starts, not after it has already cost the quarter.

2 · Freezer uptime & capacity audit

DO NEXT

Instrument freezer temperature and door-open time across all **2,443 dark stores**. Tests whether the problem is equipment and capacity rather than packaging.

WHICH LINK IT INTERRUPTS

Freezer and storage, upstream of pack-out. A freezer running warm fails before the picker ever touches the item, and no amount of packaging downstream will compensate for it.

PRIMARY METRIC

Percentage of stores operating outside the target freezer temperature band, read weekly.

WHY IT IS SECOND, NOT FIRST

This is **diagnostic, not a fix**. It tells you where liners alone will not be enough, which is valuable, but it does not by itself remove a single melted item. It also carries a capex conversation that liners do not.

WHAT IT RULES IN OR OUT

A clean freezer estate would point the whole mechanism at packaging and dwell time. A warm estate would reframe this as a capex problem rather than a consumables one. The stop rule in section 12 routes directly here.

Blinkit's 10–15 minute model relies on **dark-store cold storage, not refrigerated transit**. That makes store-side temperature the single most load-bearing assumption in the whole fulfilment chain for frozen goods.

3 · Dwell-time SLA for frozen items

THEN

In mixed baskets, frozen picks are sequenced **last** and dispatched on their own clock, so they stop queuing behind ambient items during multi-item order assembly.

WHICH LINK IT INTERRUPTS

Dwell and handling: the same exposure link as liners, but via time rather than insulation. Every minute a frozen item waits at ambient temperature during picking is a minute of cold loss.

PRIMARY METRIC

Frozen-item dwell time, p90, measured from pick to dispatch.

! GUARDRAIL

Must not degrade overall delivery time. Speed is the one dimension customers currently praise (**42,960 positive-band fast-delivery mentions**), and it should not be traded away to fix a 2.13% complaint category.

WHY IT IS LAST

It touches dispatch logic: engineering effort, and a real risk of regression against the guardrail above. It would be tested with the same treated-vs-control design as the liner pilot, with dwell-time p90 as the leading indicator.

Explicitly not recommended this quarter: rider-speed programmes, support headcount increases, cancellation-flow redesign, refund-speed initiatives. No statistically significant movement was detected in any of the categories behind them (section 05).

WHAT IS BEING RECOMMENDED, PRECISELY

Commit to a **controlled test of intervention 1**, priced before rollout, not to a national rollout, and not to capex.

A ₹3–9 gel-pack insert, tested against an illustrative ₹5.65 allowance that is itself conditional on the assumptions registered in section 10. Interventions 2 and 3 are sequenced behind the first read, and are triggered by it rather than run in parallel.

Why lowest-regret is the right frame. The attribution share is assumed, not measured, so the cost of being wrong must stay small. Liners are cheap, reversible, and confined to one arm of a pilot: the case can be wrong without Blinkit having spent much to find out.

Interventions are sequenced by cost-to-test and directness against the hypothesized mechanism (section 09), **not** by confirmed cost impact. No impact percentage is claimed for any of the three; establishing actual payback against internal cost data is the explicit purpose of the pilot in section 12. The ₹5.65 figure is a conditional spend threshold derived from two stated assumptions, not an observed procurement or cost number.

What could weaken this case, and what would change our mind.

A finding this consequential deserves an explicit account of where it could be wrong. None of these limitations were resolved by ignoring them: each is either measured, bounded, or handed to the pilot to close. The falsification criteria below are stated in advance, so the case can be checked rather than merely believed.

Limitations register			SEVERITY · AND WHAT IS BEING DONE ABOUT IT
ITEM	SEVERITY	NOTE	
Attribution share is judgement	HIGH	Bounded 10–50%, not measured. Everything downstream of it (₹18.7cr and ₹5.65) inherits that uncertainty. Internal wastage data from the pilot would replace it.	
Cluster contains contamination	MEDIUM	Quantified, not hidden. Random sampling found unrelated complaints inside the category (section 07), so 2.13% is an upper bound on pure cold-chain mentions.	
Reviewers are self-selected	MEDIUM	Mitigated by comparing composition between two equally-biased quarters rather than reading absolute levels. Mitigated, not eliminated.	
Complaint ≠ failure rate	MEDIUM	Most customers who receive a melted item never write a review. The <i>direction</i> and <i>relative magnitude</i> are informative; the absolute failure rate is not recoverable from this data.	
Seasonal baseline is thin	MEDIUM	The 2.10× expected ratio rests on two prior summers. The ≈80% / ≈20% split is an estimate against a short history, not a fitted seasonal model.	
Android only	LOW	iOS was not scraped. No reason to expect a different physical pattern, but it is untested.	
Seasonality itself	RESOLVED	Now measured directly (section 08). Roughly 80% of the rise follows the recurring summer pattern; the residual is what remains unexplained by it.	
The limitations bound the <i>magnitude</i> and <i>generalisability</i> of the finding. None of them reverses its direction , which is why the recommendation is a controlled pilot rather than a rollout.			

Confidence register			CLAIM • CONFIDENCE • WHY
CLAIM	CONFIDENCE	WHY	
Cold-chain complaint mentions rose materially QoQ	HIGH	Significant after correction, robust to version mix, replicated in an independent run.	
The seasonal pattern is real	HIGH	13-quarter back-test on one consistent category definition; present in both runs.	
Cold-chain is the leading root-cause hypothesis	MEDIUM-HIGH	Mechanism named by customers and physically plausible, but not directly observed.	
Cold-chain explains ~25% of the ₹74.6cr rate effect	LOW	A stated assumption bounded 10–50%, not a measurement.	
Liners would reduce direct cost per order	UNTESTED	No pilot has been run. This is precisely what section 12 exists to answer.	

What would change the conclusion

SEASONALITY

A **fourth consecutive summer** below the 2.10× historical baseline → this year's residual becomes noise, not signal.

MECHANISM

Internal **wastage cost flat QoQ** while complaints rose → complaints without cost following, which would undercut the mechanism entirely.

STABILITY

A **third independent run** losing cold-chain → would mean the category itself is unstable, not that the underlying effect is.

CAUSALITY

The **pilot failing to outperform control** → packaging is not the binding constraint, and the case routes to the freezer audit.

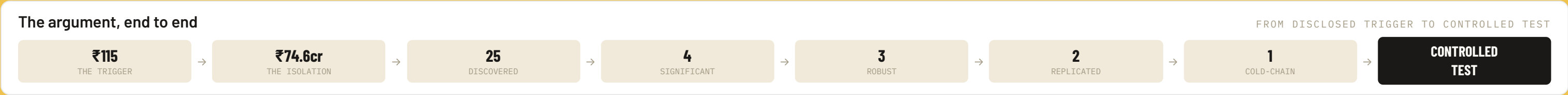
Each criterion is checkable, and three of the four can be checked without Blinkit's cooperation. Stating them in advance is what stops the case from being unfalsifiable.

FALSIFICATION CRITERIA, STATED IN ADVANCE

Limitations are listed by severity, with the highest-consequence item first. Replication evidence supporting the "stability" criterion is set out in section 06 and detailed in the supporting material; it is not restated here. No limitation in this register was discovered after the conclusion was drawn; each was identified as part of the analysis design or during the checks described in sections 05 and 06.

One quarter, one cost line, one defensible answer.

25 discovered failure modes, three independent gates, one recommendation testable in a single season. **The case concludes here**; what follows is supporting material.



Nine steps, from disclosure to decision

THE WHOLE CASE IN ONE PANEL

01 The trigger

Direct cost per order rose to ₹115, +2.0% QoQ, costing 10bps of contribution margin. Disclosed, not derived.

02 The isolation

Of ₹718cr in total increase, only ₹74.6cr is a rate effect. The rest is volume. **The rate effect is the diagnosable part.**

03 The search

25 complaint categories discovered from the corpus, with the eighth branch of the cost-driver tree left deliberately open.

04 The funnel

4 significant → 3 robust to version mix → 2 reproduced independently. Two candidates fell at later gates.

05 The selected signal

Cold-chain mentions rose 2.37×(0.90% → 2.13%), 20th of 25 by prevalence, 1st by movement, and the only survivor naming a fixable mechanism.

06 The correction

Four-fifths of that rise follows a recurring summer pattern. The residual above it is what is worth investigating.

07 The root-cause hypothesis

Freezer → pack-out → dwell → thermal exposure → degradation → wastage or refund → direct cost. One link observed.

08 The business decision

Test the lowest-regret intervention: a ₹3–9 insulated liner against an illustrative ₹5.65 allowance, stop rules written first.

09 The causal step

A matched treated-vs-control pilot plus internal wastage data: the only route from association to measured cost effect.

WHAT THIS CASE DELIBERATELY DOES NOT CLAIM

That cold-chain caused the ₹74.6cr rate rise. That the 25% attribution is measured. That the other 21 categories are irrelevant, only that no significant movement was detected this quarter. That reviews measure failure rates rather than complaint mentions.

What happens next

12 WEEKS

W0

Get the liner quote at national volume

W2

Run matched treated / control dark stores

W8

Read the gap, and price it against internal cost data

W10

Scale before next season, not after it

Ongoing

Keep cold-chain complaint share as a standing weekly measure

WHO OWNS WHAT

Procurement

Liner quote at national volume

Dark-store ops

Pack-out change, treated arm only

Analytics

Weekly complaint-share read, both arms

Finance

Wastage attribution & payback sign-off

The whole programme fits inside one quarter, ahead of the next April spike, which is the window that actually matters.

How this was produced

REPRODUCIBILITY

Data

588,832 reviews · Jul 2026

Discovery

BERTopic, unsupervised

Testing

Two-proportion z-test

Correction

Bonferroni + BH

Confound check

Version-mix reweight

Replication

Full independent re-run

Seasonality

13-quarter back-test

Power

Min detectable h = 0.040

STACK

Python

sentence-transformers

UMAP

HDBSCAN

BERTopic

statsmodels

pandas

PyArrow

WHAT A REVIEWER SHOULD CHECK FIRST

The assumption register in section 10, the gate table in section 06, and the seasonality back-test in section 08.

Stated precisely: cold-chain is the strongest validated operational signal. The pilot determines whether it is actually the cost driver.

The finding that mattered was 20th by volume and 1st by movement.

EXPLORE THE ANALYSIS: [Interactive Review Diagnostic Tool](#) · [GitHub Repository](#)

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25 → 1

CATEGORIES TO RECOMMENDATION

Appendix: the technical detail behind the gates.

▲ THE ARGUMENT ENDED ON PAGE 15

This page is **reference material, not a conclusion**, and nothing on it changes the recommendation: the replication discrepancy analysis, the clustering methodology, the statistical parameters, and the evidence-tier definitions referenced from sections 03, 05, 06, 09 and 10.

Why two independent runs disagree

RUN A VS RUN B

- 01 **UMAP is not reproducible across machines**
Even with a fixed seed, different library versions change the embedding layout, a property of the dimensionality reduction, not of the data.
- 02 **Cluster count changes the correction**
22 versus 25 categories changes the Bonferroni family size, moving borderline p-values across the 0.05 line. Produce freshness sat exactly there.
- 03 **Membership boundaries shift**
Snippets near a centroid boundary can land in either cluster, moving a share by a fraction of a point without changing what customers wrote.
- 04 **The data layer was perfectly stable**
Scraping, banding and quarter assignment produced **byte-identical cohorts** of 15,885 and 19,440 reviews in both runs. Every discrepancy originates in clustering.

Cold-chain was **first by movement in both runs** (2.68× and 2.37×), and significant and robust in both. A finding that survives a different clustering of the same reviews is not an artefact of one configuration.

Evidence tiers used throughout

HOW TO READ ANY CLAIM IN THIS DECK

TIER	MEANING	EXAMPLES IN THIS DECK
DISCLOSED	Reported by Eternal Ltd, or cross-checked coverage.	₹115/order, +2.0% QoQ, 10bps, 331M orders
DERIVED	Deterministic arithmetic on disclosed figures. No estimation.	₹74.6cr rate effect, ₹112.75 implied
MEASURED	Computed from the review corpus by our own pipeline.	2.37× movement, all 25 category shares
TESTED	A stated procedure with a pass/fail rule fixed in advance.	Significance, version mix, replication
HYPOTHEZIZED	Consistent with the evidence, but not directly observed.	Thermal / packaging / dwell mechanism
ASSUMED	A stated judgement with an explicit range; everything downstream inherits it.	25% attribution, 10% frozen-order share
UNTESTED	No evidence exists yet. Named, not implied.	Whether liners reduce direct cost per order

Every number in the main body belongs to exactly one tier. Where a figure sits downstream of an assumption, that assumption is named on the same slide.

Statistical parameters

AS APPLIED

PARAMETER	VALUE
Test	Two-proportion z-test
Cohorts	n = 15,885 vs 19,440
Snippets clustered	52,664 from 35,325
Sample floor	n ≥ 30 per category
Correction · pre-reg.	Bonferroni
Correction · expl.	Benjamini-Hochberg
Significance threshold	Adjusted p < 0.05
Power	80%, α = 0.05 two-sided
Min detectable h	0.040
Pre-registered branches	12
Gate 2 · coverage floor	≥ 50% version mix
Gate 2 · robustness rule	Sign held, ≥ ½ magnitude
Seasonality · excl.	< 3,000 neg. reviews

Tier assignment, pre-registered versus exploratory, was fixed before any test was run.

Clustering methodology

RESOLUTION SELECTION

Reviews were split into clause snippets, embedded with sentence-transformers, reduced with UMAP and clustered with HDBSCAN via BERTopic. The clustering resolution was selected from a documented resolution search against **pre-defined separation criteria**, not tuned to obtain the desired finding.

Category labels are cosmetic renamings of model-generated term lists, applied **after** the statistics were computed. No category membership was hand-assigned.

Source register

WHERE EACH INPUT COMES FROM

INPUT	SOURCE
Direct cost/order, QoQ rate, contrib. margin	Eternal Ltd Q1 FY27 results
Orders, AOV, net order value, dark-store count	Investor-deck coverage · Business Standard, Lapaas Voice, INDmoney (Jul–Aug 2026)
Review corpus	Google Play Store, Blinkit Android app · scraped Jul 2026 (Jun 2020 → Jul 2026)
Gel-pack pricing · insulated bags	IndiaMART bulk listings, Aug 2026 · TradeIndia (~₹944/unit, excluded)
Cost-driver tree, derivations, recommendations	Author's own

NOT SOURCED, BECAUSE IT DOES NOT EXIST HERE

No internal Blinkit operational data was available or used: no freezer telemetry, no wastage ledger, no store-level failure rate, no procurement quotation, and no pilot result. Every such figure is named as required-but-missing rather than estimated.

Reviews carry no store identifier, so no finding in this deck can be localised to a city, store or SKU.